

241. Bell metal is an alloy of

- (a) Cu, Zn and Sn
- (b) Cu, Zn and Ni
- (c) Cu and Zn
- (d) Cu and Sn

Answer: D — (d) Cu and Sn

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Cu and Sn matches the relevant metallurgical process or alloy property.

242. Turnbull's blue is

- (a) Ferricyanide
- (b) Ferrous ferricyanide
- (c) Ferrous cyanide
- (d) Ferri-ferricyanide

Answer: B — (b) Ferrous ferricyanide

Chemical reason: Turnbull's blue is produced when Fe^{2+} reacts with ferricyanide, historically described as ferrous ferricyanide; it gives the characteristic deep blue precipitate.

243. Addition of high proportions of manganese makes steel useful in making rails of rail-roads because manganese

- (a) Gives hardness to steel
- (b) Helps the formation of oxides of iron
- (c) Can remove oxygen and sulphur
- (d) Can show highest oxidation state of + 7

Answer: AB — (a) Gives hardness to steel and (b) Helps the formation of oxides of iron

Chemical reason: High-manganese steel is exceptionally hard and wear-resistant, so it is

suitable for railway rails and other surfaces exposed to severe abrasion and impact.

244. Copper displaces which of the metal from their salt solutions

- (a) AgNO_3
- (b) ZnSO_4
- (c) FeSO_4
- (d) All of these

Answer: A — (a) AgNO_3

Chemical reason: The correct choice is AgNO_3 . It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

245. Which of the following statement(s) is(are) correct with reference to the ferrous and ferric ions

- (a) Fe^{3+} gives brown colour with potassium ferricyanide
- (b) Fe^{2+} gives blue precipitate with potassium ferricyanide
- (c) Fe^{3+} gives red colour with potassium thiocyanate
- (d) Fe^{2+} gives brown colour with ammonium thiocyanate

Answer: BC — (b) Fe^{2+} gives blue precipitate with potassium ferricyanide and (c) Fe^{3+} gives red colour with potassium thiocyanate

Chemical reason: The correct choice is Answer: BC — (b) Fe^{2+} gives blue precipitate with potassium ferricyanide and (c) Fe^{3+} gives red



colour with potassium thiocyanate. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

246. Which of the following element constitutes a major impurity in pig iron

- (a) Silicon
- (b) Oxygen
- (c) Sulphur
- (d) Graphite

Answer: D — (d) Graphite

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Graphite matches the relevant metallurgical process or alloy property.

247. Annealing is

- (a) Heating steel in nitrogen and cooling
- (b) Heating steel to bright redness and then cooling slowly
- (c) Heating wrought iron with carbon to redness
- (d) Heating steel to high temperature and cooling suddenly by plunging in water

Answer: B — (b) Heating steel to bright redness and then cooling slowly

Chemical reason: Annealing means heating steel to a high temperature and then allowing it to cool slowly. This softens the metal and relieves internal stresses.

248. In electroplating, the metal that is not used for plating is

- (a) Fe
- (b) Zn
- (c) Ni
- (d) Au

Answer: A — (a) Fe

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability, conductivity, or corrosion resistance. That property is specifically associated with Fe.

249. Which one of the following is a wrong statement about cast iron

- (a) It is also called pig iron
- (b) It contains about 4.5% carbon
- (c) It is corrosion resistant
- (d) It contracts on cooling

Answer: D — (d) It contracts on cooling

Chemical reason: Cast iron expands slightly on solidification rather than contracting in the ordinary way, which is why it fills moulds well. Therefore the statement that it contracts on cooling is the incorrect one in the set.

250. Iron pipes lying under acidic soil are often attached to blocks of magnesium for protection



from rusting. Magnesium offers protection to iron against corrosion because it

- (a) Is more readily converted into positive ions
- (b) Is lighter than iron
- (c) Forms a corrosion-resistant alloy with iron
- (d) Prevents air from reaching the surface of iron

Answer: A — (a) Is more readily converted into positive ions

Chemical reason: Magnesium is more electropositive than iron and oxidizes preferentially. It acts as a sacrificial anode, supplying electrons to the iron and preventing iron from being oxidized.

251. FeS_2 is

- (a) Artificial silver
- (b) Fool's gold
- (c) Mohr's salt
- (d) Cast iron

Answer: B — (b) Fool's gold

Chemical reason: The correct choice is Fool's gold. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

252. Stainless steel is an alloy of iron with

- (a) 8% Cr, 50% Mn
- (b) 10% Ni, 2% Mn
- (c) 2% Cr, 3% C
- (d) 12% Cr, 1% N

Answer: A — (a) 8% Cr, 50% Mn

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in 8% Cr, 50% Mn matches the relevant metallurgical process or alloy property.

253. The chemical processes in the production of steel from haematite ore involve

- (a) Reduction
- (b) Oxidation
- (c) Reduction followed by oxidation
- (d) Oxidation followed by reduction

Answer: D — (d) Oxidation followed by reduction

Chemical reason: Haematite (Fe_2O_3) is first reduced in the blast furnace to iron. In steelmaking, excess carbon and other impurities are then oxidized, so the overall sequence is reduction followed by oxidation.

254. The protection of steel by chrome plating is due to

- (a) Cathodic protection
- (b) Anodic protection
- (c) Covering of steel surface
- (d) Formation of alloy with iron

Answer: A — (a) Cathodic protection

Chemical reason: Chrome plating protects steel mainly by physically covering the surface



with a corrosion-resistant chromium layer,
reducing contact of iron with air and moisture.

255. The most convenient method to protect bottom of ship made of iron is

- (a) White tin plating
- (b) Coating with red lead oxide
- (c) Connecting with 'Pb' block
- (d) Connecting with 'Mg' block

Answer: A — (a) White tin plating

Chemical reason: For an iron ship hull, attaching magnesium provides cathodic protection. Mg acts as the sacrificial anode and corrodes instead of the iron.

256. Carbon monoxide reacts with iron to form

- (a) $\text{Fe}(\text{CO})_5$
- (b) FeCO_2
- (c) $\text{FeO}+\text{C}$
- (d) $\text{Fe}_2\text{O}_3+\text{C}$

Answer: D — (d) $\text{Fe}_2\text{O}_3+\text{C}$

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in $\text{Fe}_2\text{O}_3+\text{C}$ matches the relevant metallurgical process or alloy property.

257. Iron is extracted from magnetite by reduction with

- (a) H_2
- (b) C
- (c) Mg
- (d) Al

Answer: B — (b) C

Chemical reason: In the blast furnace, magnetite/iron oxides are mainly reduced by carbon monoxide generated from coke; carbon can also participate indirectly. The chosen reducing agent matches this metallurgical reduction route.

258. Malachite is a mineral of

- (a) Zn
- (b) Fe
- (c) Hg
- (d) Cu

Answer: D — (d) Cu

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to Cu. The choice is fixed by the known composition of that compound/mineral.



259. The most important oxidation state of copper is

- (a) + 1
- (b) + 2
- (c) + 3
- (d) + 4

Answer: B — (b) + 2

Chemical reason: The most common and stable oxidation state of copper in aqueous chemistry is +2, corresponding to Cu^{2+} with a $3d^9$ configuration.

260. Hot and conc. nitric acid when reacts with copper, the gas obtained is

- (a) N_2
- (b) Nitrous oxide
- (c) NO
- (d) NO_2

Answer: D — (d) NO_2

Chemical reason: Hot concentrated HNO_3 oxidizes copper and is reduced mainly to nitrogen dioxide, NO_2 , a brown gas: $\text{Cu} + 4\text{HNO}_3(\text{conc.}) \rightarrow \text{Cu}(\text{NO}_3)_2 + 2\text{NO}_2 + 2\text{H}_2\text{O}$.

